Fiscal Rules with Fixed Exchange Rates and Sovereign Risk

Gastón Marinelli
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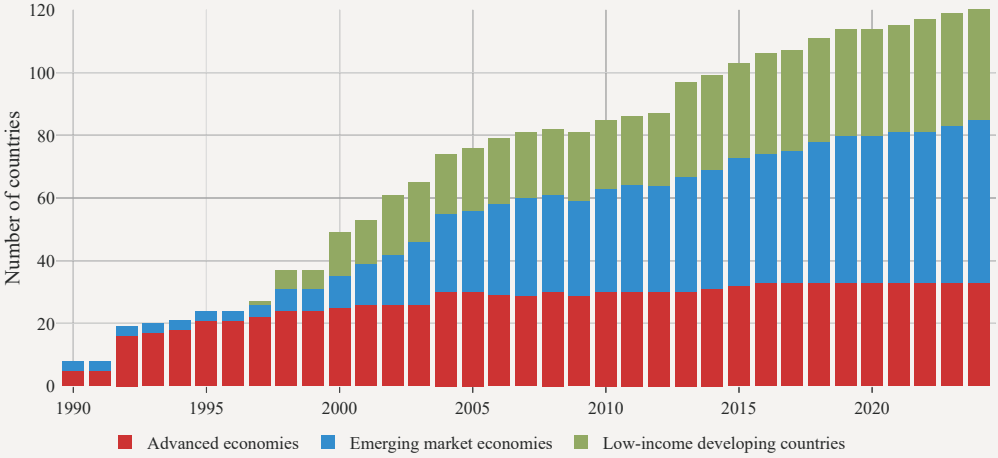
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LACEA
Lima, November 2026

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Adoption of Fiscal Rules



Source: IMF Fiscal Rules and Fiscal Councils dataset

Fiscal Rules and Fixed Exchange Rates

- Fiscal rules are an important tool to mitigate sovereign risk

- Emerging market economies differ in exchange rate regimes
  - … which affects business cycle volatility
  - … which affects behavior like reserve accumulation

How does the exchange rate regime affect the optimal design of fiscal rules?

Sources of Volatility

Main result

- Higher volatility makes tighter fiscal rules optimal

BUT

- The elasticity of tightness to volatility is not always the same
  - Volatile underlying shocks $\Longrightarrow$ strong response
  - Amplification through fixed exchange rates $\Longrightarrow$ weaker response

- We decompose the different response of tightness to volatility into
  - … conditional distribution of future income given current income
  - … fiscal multipliers

Literature Review

- Sovereign debt and default
- Fiscal rules
- Fixed exchange rates

Roadmap

- Model
- Fiscal Rules
- Rules with Flexible Exchange Rates
- A Pseudo-Fixed Exchange Rate Regime
- Conclusions

Model

Model Environment

Simplest possible TNT environment

- Stochastic endowment of tradable goods

$$\mathcal{C}(c_T, c_N) = \left(\omega_N c_N^{-\eta} + \omega_T c_T^{-\eta}\right)^{-\frac{1}{\eta}}$$

- CES demand

$$\frac{p_N}{p_T} = \frac{\omega_N}{\omega_T}\left(\frac{c_T}{c_N}\right)^{1+\eta}$$

- Nontradable goods produced with labor
  - Wage rigidities
  - Inelastic labor supply

$$w > \bar{w} \quad \text{or} \quad h < 1$$

$$w = p_N f'(h)$$

- Firm FOC gives labor demand
  - Firm's FOC + CES demand yields equilibrium employment / tradable consumption relation

$$h = \mathcal{H}(c_T, \bar{w}) = \left(\frac{\omega_N}{\omega_T}\frac{a}{\bar{w}}\right)^{\frac{1}{1+\eta\alpha}} c_T^{\frac{\eta\alpha}{1+\eta\alpha}}$$

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Government's Choices

- In repayment, gov't chooses new issuances and labor supply

$$v_R(b,z) = \max_{b'} u(c_T, c_N) + \beta \mathbb{E}\left[v(b',z') \mid z\right]$$
$$\text{subject to} \quad c_T + \kappa b = y_T(z) + q(b',z)(b' - (1-\delta)b)$$
$$c_N = f\Big(\min\{1, \mathcal{H}(c_T, \bar{w})\}\Big)$$

- In default output is reduced + no debt issuance

$$v_D(z) = u\Big(y_T^D(z), \min\{1, \mathcal{H}(y_T^D(z), \bar{w})\}\Big) + \beta \mathbb{E}\left[\psi v(0,z') + (1-\psi) v_D(z') \;\Big|\; z\right]$$
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Euler equation for issuance

- The government at (b, z) chooses new issuances b' in repayment according to

$$\left(\frac{u'(c)}{p_C} + \xi\right)\left(q + \frac{\partial q}{\partial b'}[b' - (1-\delta)b]\right) = \beta(1-\mathcal{P}')\mathbb{E}\left[\left(\frac{u'(c')}{p'_C} + \xi'\right)\right]\mathbb{E}\left[(\kappa + (1-\delta)q')\right] + \beta(1-\mathcal{P}')\text{cov}\left(\kappa + (1-\delta)q', \frac{u'(c')}{p'_C} + \xi'\right)$$

… where $\xi = \lambda\frac{\mathcal{H}'(c_T;\bar{w})}{p_T}$ is the scaled multiplier on the wage floor constraint

- Future binding constraints create precautionary behavior that fights debt dilution

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Calibration with Fixed Exchange Rate

- We calibrate to Ecuador 2010-2019
  - … after dollarization, before 2020 default
- Strategy: pick
  - Sovereign's discount factor β, default costs (d_1, d_2)
  - Wage floor $\bar{w}$, share of nontradables in consumption ω_N
  - Tradable endowment log-AR(1) parameters (ρ_z, σ_z)
- to match
  - Debt ratios, spreads
  - Unemployment, share of tradables in GDP
  - Estimation of log-AR(1) for real GDP

Parameter	Value
β	0.8989
d_1	-0.4063
d_2	0.4892
$\bar{w}$	0.7602
ω_N	0.5324
ρ_z	0.9523
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Model Fit

- Calibration to Ecuador fits reasonably well

Description	Data	Model
Average external debt-to-GDP ratio	21.8	21.8
Average spread, bps	774	759
Std. Dev. spread, bps	220	233
Unemployment rate	4.5	4.1
Autocorrelation of real GDP	0.845	0.686
Std. Dev. of real GDP	0.0121	0.0138
Share of tradables on GDP	0.42	0.48

Note: moments from pre-default samples

Fiscal Rules

Debt and Deficit Limits

We consider different constraints on the government's debt choices

- Debt limits

$$\frac{b'}{4 \cdot (p_T y_T + p_N y_N)} \leq \bar{d}$$

- Deficit limits

$$\frac{q \cdot (b' - (1-\delta)b)}{4 \cdot (p_T y_T + p_N y_N)} = \frac{p_T c_T + \kappa b - p_T y_T}{4 \cdot (p_T y_T + p_N y_N)} \leq \bar{\delta}$$

- Debt brake

$$\frac{p_T c_T + \kappa b - p_T y_T}{4 \cdot (p_T y_T + p_N y_N)} \mathbb{1}_{\left(\frac{b}{4 \cdot (p_T y_T + p_N y_N)} \geq \bar{d}\right)} \leq \bar{\delta}$$

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Evaluating Fiscal Rules

- To evaluate fiscal rules, we compute the equivalent increase in consumption from baseline

$$\int c(v_{\text{baseline}}(b,z))(1+\lambda)d\mu(b,z) = \int c(v_{\text{rule}}(b,z))d\mu(b,z)$$

- $c(v)$ is the constant consumption stream that yields value v
- $\mu(b,z)$ is the ergodic distribution of the model without rules

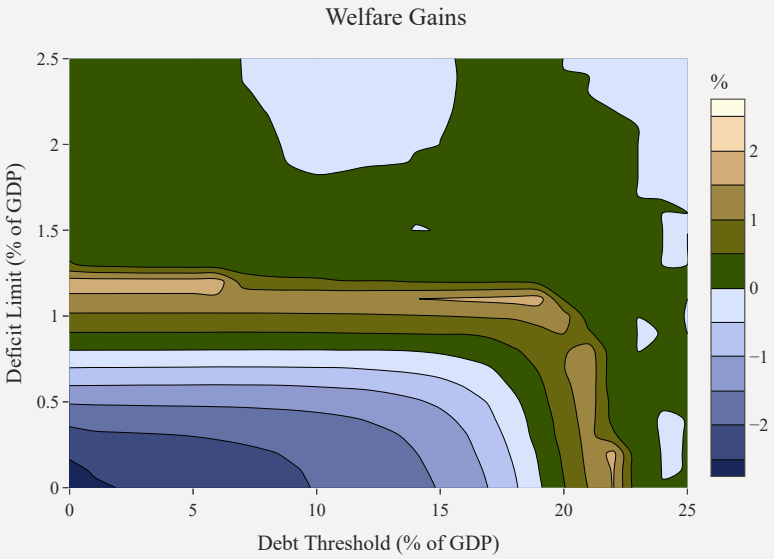
$$\Longrightarrow \quad \lambda \text{ is the equivalent increase in consumption}$$

Optimal Fiscal Rules

	Baseline	Debt Limit $\bar{d} = 22.25\%$	Deficit Limit $\bar{\delta} = 1.25\%$	Debt Brake $(\bar{d}, \bar{\delta}) = (0\%, 1.25\%)$
Debt-to-GDP (%)	20.6	21.1	21.7	21.7
Spread (bps)	670	83.1	0.34	0.34
SD. Spread (bps)	255	25.7	0.69	0.69
Unemp. (%)	3.55	2.77	2.72	2.72
Default Freq. (%)	3.04	0.33	0	0
Welfare Gain (%)	–	0.89	1.94	1.94

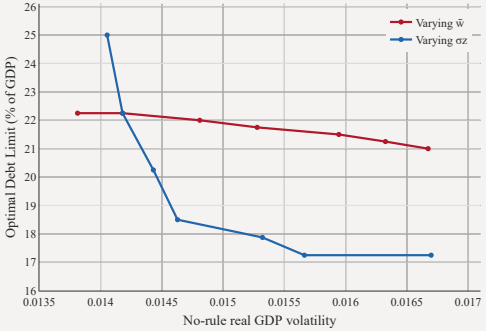
Note: Statistics from each ergodic distribution

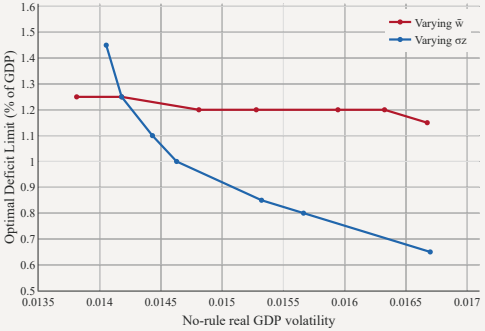
Welfare Gains of Debt Brake Rules



Debt and Deficit Limits

- Increasing volatility through σ_z tightens optimal rules much more





Rules with Flexible Exchange Rates

Environment with Flexible Exchange Rates

Unchanged TNT, but no nominal rigidities: $\bar{w} = 0$

- Calibration to the same volatility of real GDP
  - … loading on volatility of shocks

Parameter	Value
β	0.9228
d_1	-0.2164
d_2	0.3265
$\bar{w}$	0
ω_N	0.5749
ρ_z	0.9226
σ_z	0.0296

Description	Data	Model
Average external debt-to-GDP ratio	21.8	21.7
Average spread, bps	774	760
Std. Dev. spread, bps	220	225
Unemployment rate	4.5	–
Autocorrelation of real GDP	0.845	0.812
Std. Dev. of real GDP	0.0121	0.0117
Share of tradables on GDP	0.42	0.44

Note: moments from pre-default samples

Optimal Fiscal Rules with Flexible Exchange Rates

	Baseline	Debt Limit $\bar{d} = 24.5\%$	Deficit Limit $\bar{\delta} = 1.3\%$	Debt Brake $(\bar{d}, \bar{\delta}) = (0\%, 1.3\%)$
Debt-to-GDP (%)	20.7	20.5	23.4	23.4
Spread (bps)	707	455	202	202
SD. Spread (bps)	226	113	120	120
Unemp. (%)	0	0	0	0
Default Freq. (%)	3.46	2.2	0.91	0.91
Welfare Gain (%)	–	0.18	1.1	1.1

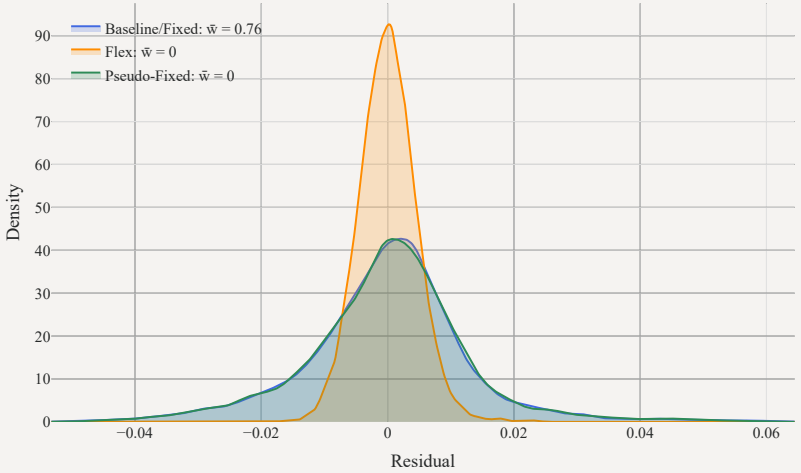
Note: Statistics from each ergodic distribution

- Optimal rule with fixed exchange rate: deficit limit $\bar{\delta} = 1.25\%$

A Pseudo-Fixed Exchange Rate Regime

Distributions of log-AR(1) Real GDP Residuals

- Same model, $\bar{w} = 0$, shock *distribution* chosen to match cond. dist. of real GDP



Optimal Fiscal Rules with a Pseudo-Fixed Model

	Baseline	Debt Limit $\bar{d} = 16.5\%$	Deficit Limit $\bar{\delta} = 0.8\%$	Debt Brake $(\bar{d}, \bar{\delta}) = (0\%, 0.8\%)$
Debt-to-GDP (%)	14.3	14.5	12.8	12.8
Spread (bps)	2151	965	0.09	0.09
SD. Spread (bps)	781	370	0.12	0.12
Unemp. (%)	0	0	0	0
Default Freq. (%)	11.8	6.52	0	0
Welfare Gain (%)	–	0.27	1.15	1.15

Note: Statistics from each ergodic distribution

- Optimal rule with fixed exchange rate: deficit limit $\bar{\delta} = 1.25\%$
- Optimal rule with flexible exchange rate: debt brake $\bar{\delta} = 1.3\%$

  … pseudo-fixed optimal rule tighter: shock distribution makes avoiding default simpler

Conclusions

Conclusions

- We study fiscal rules for economies with fixed exchange rates
- Different sources of volatility affect optimal rules differently
- Fixed exchange rate implies
  - … Different distribution of output innovations
  - … Different fiscal multipliers